

① Tabelas-verdade e expressões booleanas

1.

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

c \ AB	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$$F = C + AB$$

2.

A	B	C	G
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

c \ AB	00	01	11	10
0	0	1	0	0
1	1	1	1	1

$$G = C + \bar{A}B$$

3.

A	B	C	D	H
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

c \ AB	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	1	1
10	1	0	1	1

$$H = \bar{B} + AC$$

4.

A	B	C	I
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

c \ AB	00	01	11	10
0	0	1	0	1
1	0	1	0	1

$$I = A \oplus B$$

A	B	C	D	J
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

5.

CD \ AB	00	01	11	10
00	0	0	0	1
01	0	0	0	1
11	1	1	1	1
10	0	0	0	1

$$J = \bar{A}\bar{B} + CD$$

2) Simplificações com álgebra booleana

$$\textcircled{6} F = AB + \bar{A}\bar{B} + \bar{A}B$$

$$F = AB$$

$$\textcircled{11} F = (A + \bar{B} + C)(A + \bar{B} + C)$$

$$F = A + C$$

$$\textcircled{7} F = (A + \bar{B})(A + \bar{B})$$

$$F = A$$

$$\textcircled{12} F = AB + \bar{A}C + BC$$

$$F = AB + AC$$

$$\textcircled{8} F = \bar{A}\bar{B}C + A\bar{B}C + AC\bar{B} + ACB$$

$$F = AC$$

$$\textcircled{9} F = \bar{A}\bar{B} + \bar{A}\bar{B} + A\bar{B}C$$

$$F = \bar{A} + \bar{B}C$$

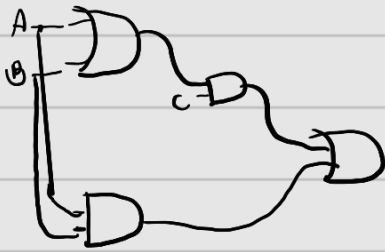
$$\textcircled{10} F = A + \bar{A}B$$

$$(A + \bar{A}) (A + B)$$

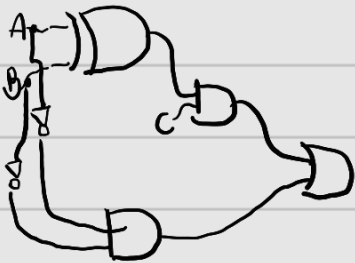
$$F = A + B$$

3) Exprimer par circuit

13) $F = (A+B)\bar{C} + AB$



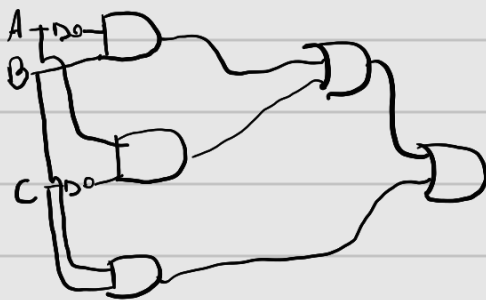
14) $F = (A \oplus B)C + \overline{AB}$



15)

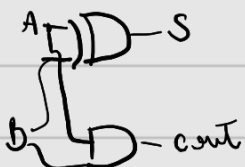


16)

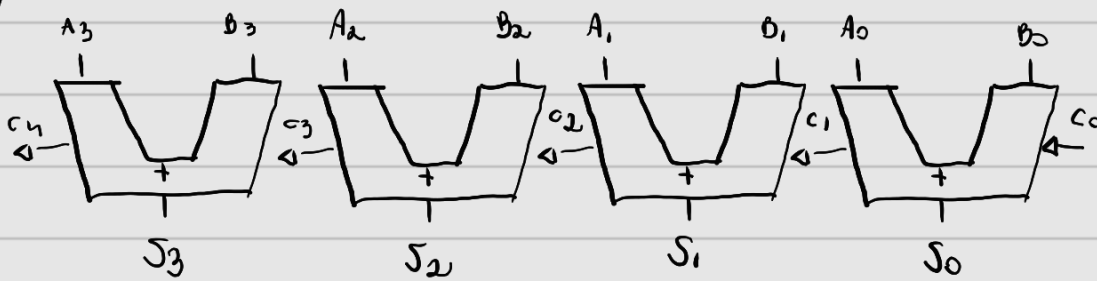


4) Somadokus

17)



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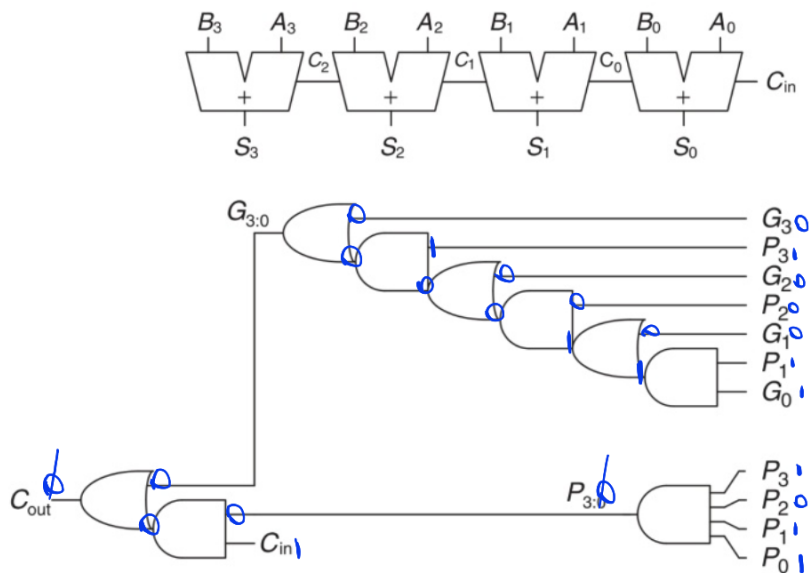
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3 2 1 0

a) 1011

+ 0001

Cin = 1

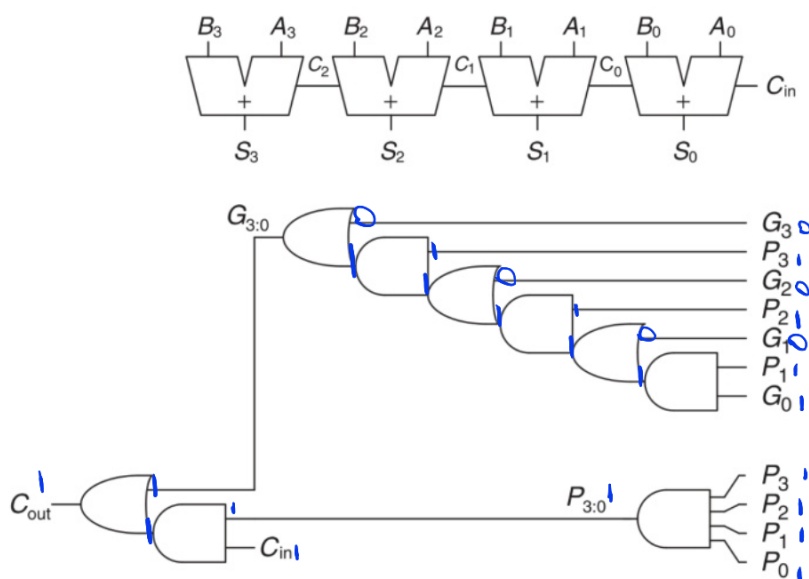


3 2 1 0

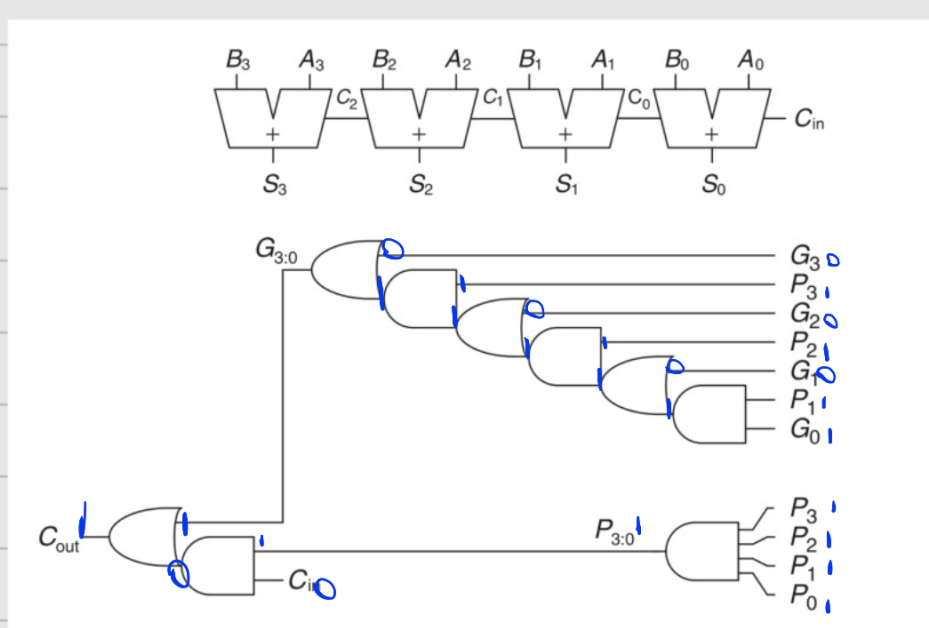
b) 1011

+ 0101

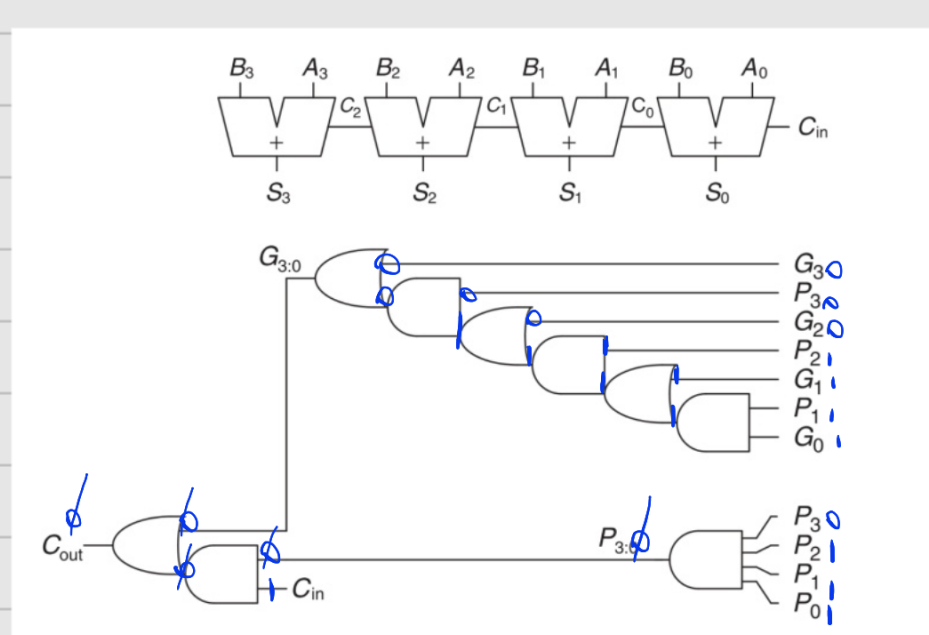
Cin = 1



3 2 1 0
 (c) 1 0 1 1
 + 0 1 0 1
 $C_{in} = 0$



3 2 1 0
 (d) 0 0 1 1
 + 0 1 1 1
 $C_{in} = 1$



(5) Circuitos para expressões

(20) $Y = A\bar{B} + A\bar{B} + AC$
 $Y = A$

(21) $Q = A\bar{B} + B\bar{C} + AC$
 $Q = A + BC$

(22) $X = A \oplus B$

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